Exploiting Specular Multipath in Automotive Radar: Virtual Aperture EKF Assimilation, Rank-1 Bearing Invariants, and Kinematics-Aware Gating

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***Abstract*—Specular radar measurements require a consistent receiving-frame transformation and a correlation-aware update. We derive the equivalence of target-image and sensor-image constructions and reproduce a 15.189-degree mismatch caused by using an unfolded ray in the wrong frame. We analyze virtual-aperture covariance, duplicate measurements, the assimilation/suppression contract, contact inversion, Hermite barriers and bounded kinematic gating. Paired instrumented replay comprises 155 observations and 10,130 assimilation marks. Those counters are distinguished from ground-truth tracking accuracy; full correlated covariance and several proposed qualifiers exceed the archived implementation.**

I Introduction and contribution boundary

Specular paths can supply geometrically useful observations, but only if the propagation path, coordinate convention and correlation structure are explicit. The engineering problem in this project is not that classical image methods are wrong. It is that a ray unfolded across a plane can be interpreted in the wrong frame at the array. We separate that frame error from the legitimate target-image construction and from the additional engineering problems of association, duplicated observations and downstream suppression. The sensor-side repository owns CFAR and CDC helpers; the Domain Controller holds the F360 tracker and replay integration. This paper analyzes a proposed virtual-aperture model and the narrower implemented hooks. It retains the archive's reported metrics while distinguishing counters from independently measured tracking accuracy.

II Related work

Kalman filtering [1], interacting multiple models [2], and adaptive CFAR [3] provide the classical estimation and detection background. A specular plane admits an image-source construction; using an image sensor is a coordinate choice, not evidence that all target-image models have an inherent bearing error. The project-specific issues are a covariance-aware observation contract, implementation ordering and reproducible replay accounting. Patent disclosure drafts in the appendix express possible embodiments rather than a completed prior-art or patentability assessment.

III Reflection geometry and the bearing-frame mismatch

Let the reflector be $\mathbf{n}^T\mathbf{x} = d$, where $\|\mathbf{n}\| = 1$. Define the Householder linear map $H = I - 2\mathbf{n}\mathbf{n}^T$ and affine reflection

$$F(\mathbf{x}) = H\mathbf{x} + 2d\mathbf{n}. \tag{1}$$

The actual sensor is $\mathbf{s}$, the target $\mathbf{p}$, and their images are $\mathbf{s}^* = F(\mathbf{s})$ and $\mathbf{p}^* = F(\mathbf{p})$. Then

$$\mathbf{p}^* - \mathbf{s} = H(\mathbf{p} - \mathbf{s}^*). \tag{2}$$

Since $H^T H = I$, both rays have exactly equal norms. Their angle coordinates differ unless the ray is first transformed back into the receiving frame. This is an equivalence between correct image methods, not a contradiction in geometric optics.

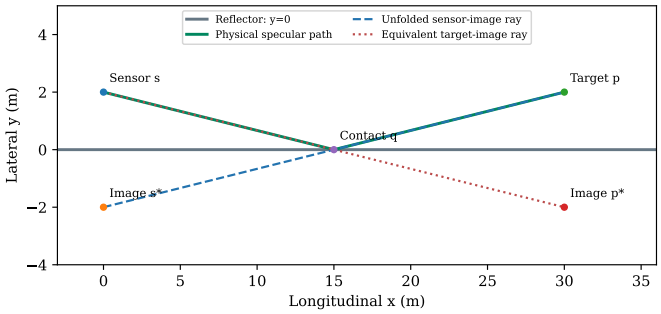


Figure 1: Explicit plane-reflection example. Solid green is the physical path; the two unfolded image constructions are equivalent after Householder transformation.

III-A A reproducible numerical example

Choose wall $y = 0$, sensor $(0,2)$ m and target $(30,2)$ m. The contact point is $(15,0)$ m, sensor image $(0,-2)$ m and target image $(30,-2)$ m. The ray from the sensor toward the contact has direction $(30,-4)$ while the untransformed image-sensor ray is $(30,4)$. Their signed angles are $\mp\arctan(4/30)$, so confusing them produces

$$\Delta\theta = 2\arctan(4/30) = 15.1893^\circ. \tag{3}$$

The apparent ray endpoints at longitudinal coordinate 30 m differ by 8 m laterally. This is not a general error at every 30 m slant range. A mixed direct/reflected round-trip path also

requires half the total propagation distance for a conventional monostatic range readout; it is not universally equal to the one-way image-ray norm.

III-B Rank-one displacement: a precise definition

For an origin-centred sensor, $\mathbf{s}^* = 2d\mathbf{n}$. Define a displacement coordinate $\mathbf{p}_s = \mathbf{p} - \mathbf{s}^*$, giving

$$\mathbf{p} - \mathbf{p}_s = 2d\mathbf{n}. \tag{4}$$

Columns sharing the same displacement direction lie in a rank-one subspace. If $\mathbf{p}_s$ is constructed by this definition, the residual is zero identically and cannot validate an independent correspondence. A useful residual requires independently inferred pair geometry. For a conventional mirrored target, $\mathbf{p} - F(\mathbf{p}) = 2(\mathbf{n}^T\mathbf{p} - d)\mathbf{n}$: direction is shared, but magnitude generally varies with target position.

IV Virtual-aperture EKF and coupled covariance

For a target state $\mathbf{x}$ and uncertain reflector parameters ξ, stack direct and specular observations:

$$\mathbf{z} = \begin{bmatrix} h_d(\mathbf{x}) \\ h_g(\mathbf{x}, \xi) \end{bmatrix} + \nu. \tag{5}$$

Linearizing in ξ around its estimate gives the marginal observation covariance

$$R_{\text{eff}} = R_0 + J_\xi \Sigma_\xi J_\xi^T. \tag{6}$$

Here J_ξ maps parameter perturbations into measurement units and Σ_ξ has corresponding squared parameter units. Shared reflector terms can create off-diagonal correlations across observations. Positive measurement noise plus positive-semidefinite reflector covariance yields a positive-definite effective covariance.

$$S = H_x P H_x^T + R_{\text{eff}}, \quad K = P H_x^T S^{-1}. \tag{7}$$

$$P^+ = (I - KH_x)P(I - KH_x)^T + K R_{\text{eff}} K^T. \tag{8}$$

The Joseph covariance form guards symmetry and positive semidefiniteness under finite precision. Inverse notation describes the mathematics; numerical implementations should solve a factored system rather than explicitly forming the inverse.

IV-A Range and bearing Jacobians

For displacement $(u_x, u_y) = \mathbf{p} - \mathbf{s}^*$, define $\rho = \sqrt{u_x^2 + u_y^2}$ and $\theta = \operatorname{atan2}(u_y, u_x)$. Their target-position Jacobian is

$$J_p = \begin{bmatrix} u_x/\rho & u_y/\rho \\ -u_y/\rho^2 & u_x/\rho^2 \end{bmatrix}. \tag{9}$$

Reflector distance and normal angle affect $\mathbf{s}^*$ and must be propagated through this Jacobian. A universal scalar factor of two does not capture all range/bearing units. Near-zero range, grazing contacts and poorly known walls require rejection or explicit robustification.

IV-B Implemented scope and duplicate-observation risk

The archived helper implements scalar variance inflation $r_{\text{base}} + (j\sigma_\xi)^2$, not the complete correlated matrix. The CCA update appends a ghost range-rate row when a selected candidate passes a Doppler consistency check; the CTCA path inflates an existing row instead. These paths are different embodiments. If an appended observation has already contributed to an existing row, diagonal noise alone does not prevent double counting. For two identical scalar observations with covariance $\sigma^2 \left[\begin{smallmatrix} 1 & c \\ c & 1 \end{smallmatrix}\right]$, their combined information is $2/[\sigma^2(1+c)]$, which tends to the information of one observation as $c \to 1$. Treating them as independent exaggerates information.

V Assimilation versus suppression: the XOR contract

The out-of-band value `APRISM_ASSIMILATED_MIRROR_PROB` equals -1.0 and marks a track that consumed an observation in the current scan. Penalty sites consult this state and skip incompatible suppression. Scan close-out resolves it to zero. The contract excludes simultaneous assimilation and penalty; it does not imply that every unassimilated candidate must be penalized. The per-detection sidecar is a bounded static array and its lifecycle must be reset per scan. Index reuse across objects, sensors or invocations needs explicit ownership. The archive reports a uint8 log value of 157, but a generic float-to-unsigned conversion outside the representable range is not a portable modulo specification. For a signed integer -100, modulo 256 would be 156, not 157. Resolving the sentinel and clamping before conversion is the appropriate repair; the specific historical byte needs its original reproducer.

VI Contact inversion and curved barriers

Let $\mathbf{u}$ be a unit sensor-to-contact direction, $\delta = \mathbf{p} - \mathbf{s}$, $\mathbf{q} = \mathbf{s} + t\mathbf{u}$, and $\ell = t + \|\delta - t\mathbf{u}\|$ the one-way broken-path length. Squaring $\ell - t = \|\delta - t\mathbf{u}\|$ gives

$$t = \frac{\ell^2 - \|\delta\|^2}{2(\ell - \mathbf{u}^T\delta)}. \tag{10}$$

Require positive denominator, $0 < t < \ell$, and a nondegenerate reflected direction. This is not a circle of fixed radius ℓ: the residual leg has length $\ell - t$. If the input is monostatic reported range rather than total broken-path length, its conversion must follow the propagation model. For two contact positions and slopes, a cubic $y(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ satisfies a confluent Vandermonde system:

$$\begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 \\ 0 & 1 & 2x_1 & 3x_1^2 \\ 1 & x_2 & x_2^2 & x_2^3 \\ 0 & 1 & 2x_2 & 3x_2^2 \end{bmatrix} \mathbf{c} = \begin{bmatrix} y_1 \\ m_1 \\ y_2 \\ m_2 \end{bmatrix}. \tag{11}$$

Its determinant is $(x_2 - x_1)^4$. Distinct abscissae establish algebraic uniqueness, not good numerical conditioning. A shifted/scaled local coordinate improves conditioning. This is a Hermite cubic approximation, not an exact clothoid, whose

curvature is linear in arc length rather than in a monomial graph coordinate.

VII Doppler confirmation and kinematic gates

A Doppler candidate can be admitted when

$$|v_r - \mathbf{u}^T \hat{\mathbf{v}}| \le 3\sqrt{\sigma_r^2 + \mathbf{u}^T P_v \mathbf{u}}. \tag{12}$$

The variance terms are in m^2/s^2 and must be nonnegative. Confirmation across two scans requires persistent pair identity and explicit scan epochs. The archived det-indexed helper can count repeated calls rather than truly consecutive scans if invoked more than once per pair, and it clears after a success. This is an integration limitation to test, not proof of persistent track identity.

VII-A Bounded adaptive gating

The proposed dimensionless threshold policy is

$$\gamma = \mathrm{clip}_{[9.21,40]}\left[9.21(1 + 0.8a_n + 0.5e_n)\right], \tag{13}$$

where $a_n = \|\mathbf{a}\|/a_{\max}$ and $e_n = \|\mathbf{y}\|^2/\mathrm{tr}(S)$ only when innovation components have compatible units. Otherwise whiten the innovation and use $\mathbf{y}^T S^{-1}\mathbf{y}$. The value 9.21 is the 99th percentile for two chi-square degrees of freedom, not three. Once a threshold depends on the candidate innovation, the original fixed-gate coverage guarantee no longer follows without calibration. An IMM can predict maneuver probabilities before association through its transition model. A narrow shared association gate can nevertheless starve a maneuver mode of evidence. This is an implementation-specific onset limitation, not a universal impossibility result for IMM tracking [2]. The archived production call normalizes acceleration by 3 m/s^2 and supplies zero innovation energy at that site; the general two-term formula is broader than that call. Its frozen rank gate remains neutral when its version is zero, so the proposed order-statistic condition $S_{(231)} \le q \le S_{(247)}$ is not a calibrated production result.

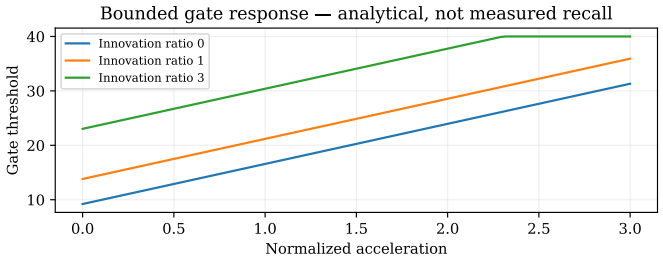


Figure 2: Analytical gate policy and saturation. This plot does not represent an observed cut-in recall curve.

VIII Environmental robustness and sensor front end

For normalized spectral powers p_i, entropy $H = -\sum_i p_i \log p_i$ can scale covariance as

$$R(t) = R_0 \exp\left(\beta H / \max(\mathrm{SNR}, \varepsilon)\right). \tag{14}$$

SNR is linear, β is dimensionless and $\varepsilon > 0$ prevents division by zero. The implementation floors SNR but does not bound exponent growth; finite inputs, overflow behavior and saturation need qualification. The WSE feed defaults to zero entropy, yielding identity scaling until an estimator supplies measurements. The reported spray-RMSE reduction is not demonstrated by that identity path. The dual-loop CFAR helper updates a range-bin clutter floor using an EMA of eligible-cell median. Occupancy-BLUE weights depend on contamination hypotheses; the current overlay supplies all-free occupancy, so a full tracker-conditioned adaptive detector is a proposed extension rather than an established effect. Median-to-mean conversion must match the exponential-noise model and quantization units.

$$f_{t+1}(r) = f_t(r) + \lambda\left[\mathrm{median}(z_{r,i}) - f_t(r)\right]. \tag{15}$$

Here $\lambda = 0.02$ in the archived defaults and floor and samples must share units. The CDC score $s_i = \mathrm{SNR}_{i,\mathrm{lin}}/r_i^2$ prioritizes strong nearby returns in a global formulation. The current integration reorders inside a range bin, where r_i is constant, and preserves count. It therefore does not demonstrate globally optimal urban bandwidth allocation or removal of detections.

IX Experimental evidence and its limits

The strongest directly inspectable artifacts are the paired CSV outputs of an instrumented replay harness. They contain scan identifiers, track and detection counts, association counts, assimilation marks and a sum of longitudinal track velocities. They do not contain independently surveyed target trajectories, classification labels, position errors or recall denominators. Accordingly, changes in these counters are observations about the harness, not estimates of fleet accuracy or false-alarm probability. The harness additionally constructs pairing candidates using azimuth and range proximity; such labels are hypotheses rather than truth annotations. The association-count difference

Table 1: Paired replay ledger, independently summed from CSV files

Quantity	OFF	ON
Replay records	155	155
Valid detections	71,779	71,779
Association events	9,478	9,131
Assimilation marks	0	10,130
Records with marks	0	151

is -347, or -3.66%. It is not a verified reduction in false associations, because the CSV files do not distinguish correct and incorrect matches. Marks may be counted at an instrumented update site and should not be equated automatically to distinct reflected targets. Repeated targets and repeated use of a detection must be audited separately. The recorded mean and maximum longitudinal velocity shifts, 0.910 and 3.743 m/s, are changes relative to OFF, not velocity errors relative to truth. The archive reports parent position RMSE 1.969 to 0.651 m, spray RMSE 0.911 to 0.315 m, fixed/adaptive cut-in recall 86.3/99.6%, and IMM recall 88.0%. Those figures are retained

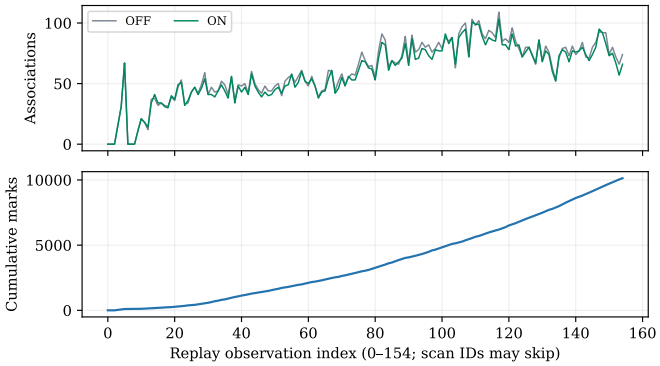


Figure 3: Actual paired replay association series and cumulative ON assimilation marks. No random series or interpolated per-scan observations are used.

as reported benchmark claims. Their truth trajectories, sample definitions and executable scoring pipelines are not provided by the paired CSVs. They are therefore not demonstrated fleet accuracy results in this manuscript. The reported 67.0% parent-RMSE reduction is the arithmetic $1 - 0.651/1.969$; correct arithmetic does not validate the underlying experiment.

IX-A Host checks and module-level evidence

The archival ledger records 40/40 host checks for both configurations. Equal pass counts establish that the named checks passed; they do not establish binary equivalence, absence of undefined behavior, timing safety, or functional safety. A strict baseline regression comparison requires matching inputs, reset states, compiler options, output fields and tolerances. The four module probes report identity fill (T1), entropy inflation (T2), assimilation marking (T3), and penalty exclusion (T4). In T2, $0.5\exp(2) = 3.694528$ is a deterministic arithmetic probe, not a wet-road accuracy test.

IX-B Validation design required for a publication claim

Partition complete drives before parameter tuning. Keep sensor poses, calibration history and overlapping temporal windows within a single split to avoid leakage. Establish reflector and target ground truth independently of the algorithm under evaluation. Report position and velocity errors in common coordinates, per-target association precision and recall, and duplicate-update rates. Aggregate confidence intervals at drive or target-trajectory level rather than pretending adjacent scans are independent. Publish ablations for gating, reflector covariance, confirmation, seeding and geometry separately. A hardware deployment claim additionally needs the production DSP build and measured worst-case execution time.

X Conclusion

The project exposes useful estimator and interface design problems, but reproducibility requires a strict separation between equations, implemented call sites and measured evidence. The source supports inspection of guarded kernels and replay counters; it does not by itself support universal lock-time, safety or fleet-accuracy guarantees. The next decisive experiment is an independent ground-truth evaluation with complete configuration and per-drive ablations.

References

[1] R. E. Kalman, “A New Approach to Linear Filtering and Prediction Problems,” Journal of Basic Engineering, vol. 82, pp. 35–45, 1960, doi:10.1115/1.3662552.

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[3] H. Rohling, “Radar CFAR thresholding in clutter and multiple target situations,” IEEE Transactions on Aerospace and Electronic Systems, vol. AES-19, pp. 608–621, 1983, doi:10.1109/TAES.1983.309350.

[4] S. M. Kay, “A fast and accurate single frequency estimator,” IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. 37, pp. 1987–1990, 1989, doi:10.1109/29.45547.

[5] A-PRISM project, “Master archival backup,” one.md, October 2026. Internal source/equation/claim ledger; not an independently reviewed publication.

[6] A-PRISM project, paired instrumented replay CSVs, real_off.csv and real_on.csv. Source files and hashes reproduced in the extended reproducibility appendix.

APPENDIX A: INVENTION DISCLOSURE FORM (1 OF 2)

Status: internal technical disclosure draft; inventors, assignee and filing details are not supplied.

Field of invention

This disclosure concerns specular radar tracking and measurement lifecycle management in automotive radar software and embedded signal processing.

Background and failure modes

State continuity, coordinate consistency and observation ownership can fail independently. An estimator that appears converged can remain biased; a correct image construction can be consumed in the wrong coordinate frame; a measurement can be counted twice or penalized after assimilation. Existing estimation and image-method techniques are background knowledge, not asserted absent from prior art.

Detailed embodiment

The embodiment combines explicit coordinate and unit conventions, bounded state transitions, provenance-qualified inputs and instrumentation at the production interface. The manuscript defines the mathematical operations; the reproducibility appendix identifies implemented helpers and integration gaps. Proposed checks are not represented as already present in every archived source path.

Commercial detectability analysis

A controlled reset/replay comparison can observe angle-state transients, seed acceptance, measurement ownership and covariance changes. Telemetry alone does not uniquely identify an internal algorithm: alternative designs can produce similar curves. Source inspection, configuration records and reproducible boundary tests are stronger evidence than a single output trace.

Evidence and disclosure boundaries

Reported performance values require their original scoring data. This document does not establish legal novelty, claim validity, infringement or a safety certification. The claim draft deliberately avoids promising a universal lock time or a particular accuracy gain. Inventorship and disclosure dates must be completed from actual project records.

APPENDIX A: CLAIMS SPECIFICATION (2 OF 2)

One independent claim followed by eight dependent claims.

Claim 1 (independent) A method of radar tracking in the presence of a specular reflector, comprising identifying a candidate reflected path under an explicit receiving-frame transformation, forming a target observation with reflector-parameter uncertainty, applying a covariance-aware state update, recording a scan-scoped assimilation state, and excluding a conflicting suppression operation for that same assimilated observation within the scan.

Claim 2 (dependent) The method of claim 1, wherein the receiving-frame transformation uses a Householder reflection across a calibrated plane.

Claim 3 (dependent) The method of claim 1, wherein reflector uncertainty produces cross-observation covariance terms.

Claim 4 (dependent) The method of claim 1, wherein duplicate use of a detection is excluded or modeled with correlated measurement noise.

Claim 5 (dependent) The method of claim 1, wherein assimilation is represented by an out-of-band sentinel resolved before serialization.

Claim 6 (dependent) The method of claim 1, wherein serialized probability fields are clamped into a defined representable range.

Claim 7 (dependent) The method of claim 1, wherein path confirmation uses persistent pair identifiers and separate scan epochs.

Claim 8 (dependent) The method of claim 1, wherein a bounded association threshold depends on predicted acceleration and a dimensionless innovation statistic.

Claim 9 (dependent) The method of claim 1, wherein reflector contact position and tangent constraints define a locally conditioned Hermite cubic approximation.